MODULE CODE: GENMP:402

MODULE: APPLY MECHANICS AND PROPERTIES OF MATTER

RQF Level: 4

Elements of Competency and Performance Criteria

Elements of competency	Performance criteria
1. Describe laws of motion and their applications	1.1. The concept of a force is clearly interpreted based on types of forces
	1.2. Newton's laws of motion are clearly interpreted based on mathematical expressions
	1.3. Newton's laws are properly applied based on the free body diagram
2. Apply static equilibrium and elasticity	2.1. The equilibrium conditions are correctly explained for rigid bodies
	2.2. Static equilibrium is properly applied base on free body diagram
	2.3. Elastic properties are properly described based on the type of solid material
3. Analyze fluid mechanics	3.1. The concept of pressure is correctly described based on its variation with depth
	3.2. Archimedes' principle is properly applied based on buoyant forces
	3.3. Fluid dynamics are effectively analysed based on ideal fluid model
4. Apply thermodynamics	4.1. Fundamental concepts of thermodynamics are clearly described based on system
	4.2. Thermodynamic laws are properly explained based on the system
	4.3. Applications of thermodynamic laws are correctly discussed based on area of application.
5. Examine effects of electric current flow in DC electric circuit	5.1. Simple electric circuit is properly described based on ohm's law
	5.2. Electric current, resistances and voltages in DC electric circuits are accurately determined based on Kirchhoff's laws
	5.3. Electric energy, work and power in DC electric circuit are correctly determined in accordance with the law of conservation of energy
6. Apply geometric instruments	6.1. Optical instruments are effectively described based on its corresponding types
	6.2. Magnification of optical instruments is correctly determined in accordance to the image location
	6.3. Optical aberrations are effectively described based on their types

Learning unit 1: DESCRIBE LAWS OF MOTION AND THEIR APPLICATIONS

LO1. Describe the concept of the force

1.1.1 Contact force and field forces (non-contact force)

Force is a pull or push upon an object.

-Contact force: exists when two objects are in contact with each other. It is a direct force and requires a contact.

Examples: friction force, tensional forces, normal force, air resistance force, applied forces, spring forces...

- Field force (non-contact force): is a force which does not require direct contact between the bodies. In this case, the objects interact without touching each other.

Examples: magnetic forces, gravitational forces, and electrical forces.

1.1.2. Fundamental forces and their characteristics

Fundamental forces: are the interactions that do not appear to be reducible to more basic interactions. These forces govern everything that happens in the universe.

There are four (4) fundamental forces:

1. Gravitational forces
2. Electromagnetic forces
3. Weak forces
4. Strong forces

Fundamental forces are characterized on the basis of the following four criteria:

- **Strength**: How strong the force is
- **Range**: How far away the force can act
- **Particles affected**: The types of particles that experience the force
- **Particles that mediate the force**: The particles that transmit the force

1.1.3. Internal and external forces

a. Internal forces: are produced from the external forces acting on structure members.

such as poled, beamed or columned. Generally, we have three types of internal forces: axial, shear, and moment.

1. Axial force: sometimes is called normal force, is compression or tension force acting aligned with expression of structure member.

2. Shear force: is force acting in a direction perpendicular to alignment of member.

3. Moment force: lastly, is turning result of force multiplied by distance from its acting location to turning point.

b. External force: it is acting on a system from outside the system as opposed to internal forces which act between components within the system.

Examples of external forces: applied force, normal force, friction force, and air resistance force.

LO1.2. Interpret Newton's laws

1.2.1 First law of Newton (law of Inertia)

Newton's first law states that "A body will remain at rest or moving at constant speed in the same direction unless it is acted upon by a force."

Examples of Newton's first law:

1. An object placed on a plane surface
2. A ball rolling on ground
3. Shaking a tree

1.2.2. Second Newton's law

Newton's second law states that "the acceleration of an object is directly proportional to the net force acting on it and inversely proportional to its mass."

This law is mathematically represented as follows;

$$a \propto \frac{F}{m}.$$

This means, $a = k\frac{F}{m}$ where k is a constant.

Experiments have shown that the value of k =1. Hence,

$$a = \frac{F}{m}.$$

By cross multiplication we get $F = ma$

F: force

A: acceleration

M: mass

1.2.3. Newton' third law of motion

Newton's third law states that "To every action there is an equal and opposite reaction."

Examples: 1. A book resting on a table applies a downward force equal to its weight on the table.

2. A person pushes against a wall (action force), and the wall exerts an equal and opposite force (reaction force) against the person.

LO1.3. Apply Newton's laws

1.3.1. Motion on a plane

Motion on a plane is the point where we consider motion in two dimensions as only two dimensions make a plane. In this case we consider both directions (X-axis and y-axis).

Equations of motion in a plane

Apply equations of motion in straight line separately in both directions, X and y.

$$V_x = U_x + a_x t \qquad V_y = U_y + a_y t$$

$$S_x = U_x t + \frac{1}{2} a_x t^2 \qquad S_y = U_y t + \frac{1}{2} a_y t^2$$

$$V_x{}^2 = U_x{}^2 + 2a_x s \qquad V_y{}^2 = U_y{}^2 + 2a_y s$$

Example: A motorboat of 40kg starting from rest on a lake accelerates in straight line. It travels a distance of 96m in 8sec.

i)Calculate it acceleration.

ii) By applying Newton's 2nd law of motion, calculate the force of the motorboat.

STATIC AND DYNAMIC(KINETIC) FRICTION

A. **Static friction:** is a friction present between two or more objects that are not moving with respect to other. EX: A book on a table.

Laws of static friction

There are two laws of static friction:

- ✓ First law: The maximum force of static friction is not dependent on the area of contact.
- ✓ Second law: The maximum force of friction is comparative to the normal force. I.e. if normal force increases, the maximum external force that can endure without moving, also increases.

Note that, static friction force, F_s is directly proportional to the normal force, N

$$\boldsymbol{F_s \propto N}$$

$$\boldsymbol{F_s = \mu_s N}$$

Where $\boldsymbol{F_s}$ is static friction force, $\boldsymbol{\mu_s}$ is coefficient of static friction force, **N** is normal force and is given by $\boldsymbol{N = mg}$, **m** is mass and **g** is acceleration due to gravity.

- ✓ **Coefficient of friction** is a ratio of friction force and normal force. Therefore, coefficient of static friction force, $\boldsymbol{\mu_s}$ is given by: $\boldsymbol{\mu_s = \frac{F_s}{N}}$
- ✓ Coefficient of friction is **dimensionless.**
- ✓ Coefficient of friction helps an object to lie on a surface.

B. **Kinetic friction:** is a friction a force that acts between moving surfaces and it is denoted as $\boldsymbol{Fk}$.

Laws of kinetic/dynamic friction

There are four (4) laws of kinetic friction:

1st law: the force of kinetic, $\boldsymbol{F_k}$ is directly proportional to the normal reaction, **N** between two surface in contact where $\boldsymbol{\mu_k}$ is constant called coefficient of kinetic friction.

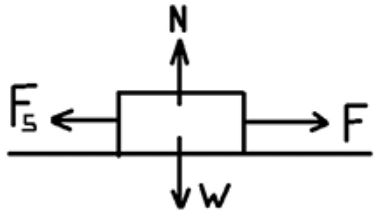
$$\boldsymbol{F_k \propto N}$$

$$\boldsymbol{F_k = \mu_k N}$$

2nd law: Force of kinetic friction is independent of shape and apparent area of the surfaces in contact.

3rd law: Force of kinetic friction depends upon nature and material of surface in contact.

4th law: Force of kinetic friction is independent of the velocity of object in contact provided the relative velocity and surface is not too long.



Difference between static and kinetic friction force

Static friction	Kinetic friction
Static fiction is a friction present between two or more objects that do not moving respect to each other.	Kinetic friction is the friction present between two or more objects that are in motion with respect each other.
The magnitude of static friction is greater than due to the greater value of its coefficient.	The magnitude of kinetic friction is comparatively lesser due to the low value of its coefficient.
The equation representing static friction is given by: $F_s = \mu_s N$	The equation representing kinetic friction is given by: $F_k = \mu_k N$

Examples

1. An object having a mass of 10kg is placed on smooth surface. Static friction between these two surfaces is given as 15N. Find the coefficient of static friction.

2. the normal force and static frictional force of an object are 50N and 80N respectively. Find the coefficient of static friction.

3. What is the relationship between static and kinetic friction? **Ans: The force of static friction keeps a stationary object at rest. Once the force of static friction is overcome, the force of kinetic friction is what slows down a moving object.**

4. A fridge weighs 1619N and the static friction coefficient is 0.50. what is the least force used to move the fridge?

5. A worker exerts a force of 400N on a large cardboard of mass 75.0kg with a coefficient of kinetic friction of 0.520.

a) Find the magnitude of force of friction.

b) Calculate the net force moving the cardboard.

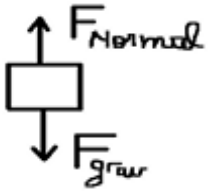
FREE BODY DIAGRAM ON A PLANE

Free body diagrams: are diagrams used to show the relative magnitude and direction of all forces upon an object in a given situation.

Rules of drawing free diagram

1. Draw cycle or square around the object of interest to be sure you focus on labeling the forces acting on an object If you are treating object as particle (no size or shape and no rotation), represent the object as point.
2. Include all forces that act on the object, representing this force as vectors. Consider the types of forces (normal force, friction, tension, and spring forces as well as weight and applied force). Do not include net force on the object. Also do not include action and reaction pair.
3. Convert the free body diagram into more detailed diagram showing the X- and y- component of a given force. This helps to solve a problem using 1st or 2nd law of Newton.
4. If there are two or more objects or bodies in the problem, draw a separate free body diagram for each object.

Example: A book is at rest on tabletop. Draw the forces acting on the book.
Ans:



1.3.2. Motion of suspended object

Suspended object/body: is an object/body hung so as to be free on all side except at the point of support.

A. Weight of suspended body

Suspended body is not in motion, its weight is due to the gravitational acceleration, **g.** so weight of suspended body/hanging body is product of its mass and gravitational acceleration.

$$W = mg$$

Where m is mass, g is gravitational acceleration ($g=9.8m/s^2$), and W is weight.
The SI unit of weight is **Newton (N)** or **kgm/s^2.**

Ex: A hanging spring balance whose 50kg was measuring the sack of 100kg. Find the weight of spring balance and sack.

Ans: let m_t be total mass of balance and sack and W be weight of sack and balance

$$m_t = 50kg + 100kg = 150kg$$

Then, $W = m_t g$

$$W = 150kg \times 9.8m/s^2$$

$$W = 1470N$$

B. TENSION FORCE

- ✓ **Tension force** is the force generated when a load is applied at one or more ends of material in directional away.
- ✓ It is often given as pulling force.

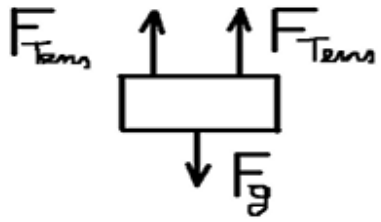
Examples:

- ✓ Pulling a block with the help of a rope
- ✓ Crane machine
- ✓ Gym equipment

FREE BODY DIAGRAM OF SUSPENDED OBJECT

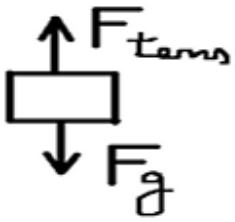
Example1: A gymnast holding onto a bar is suspended motionless in mind-air. The bar is supported by two ropes that attach to ceiling. Diagram the forces acting on the combination of gymnast and bar.

Ans: A free diagram for this situation looks like this:



Example2: a college student rests a backpack upon his shoulder. The pack is suspended motionless by one strap from one shoulder. Draw a free diagram for this situation.

Ans: A free diagram for this situation looks like this:



C. Rocket motion

Rocket motion: is based on Newton's 3rd law which states that for every action there is an equal and opposite reaction.



u: exhaust velocity, V: Rocket velocity.

❖ **Trust**

✓ **Trust** is a force which moves the rocket through the air and through space.

✓ It is given by:

$$\boldsymbol{T = v_e \frac{\Delta m}{\Delta t}}$$

Where $\boldsymbol{T}$ is trust force, $\boldsymbol{v_e}$ is escape velocity or exhaust velocity, $\frac{\Delta m}{\Delta t}$ is fuel burn rate.

✓ Acceleration of rocket is given by:

$$\boldsymbol{a = \frac{v_e}{m}\frac{\Delta m}{\Delta t} - g}$$

Where **a** is acceleration of rocket, **m** is mass of rocket, $\boldsymbol{v_e}$ is escape velocity or exhaust velocity, $\frac{\Delta m}{\Delta t}$ is fuel burn rate and **g** is acceleration due to gravity.

Example: A Saturn V's mass at liftoff was $2.8 \times 10^6 kg$, its fuel burn rate was $1.4 \times 10^4 kg/s$ and the exhaust velocity was $2.4 \times 10^3 m/s$. Calculate its initial acceleration.

Ans: Given: $\boldsymbol{m = 2.8 \times 10^6 kg, \frac{\Delta m}{\Delta t} = 1.4 \times 10^4 kg/s, v_e = 2.4 \times 10^3 m/s}$ **and**

$$\boldsymbol{g = 9.8 m/s^2}$$

Formula: $\boldsymbol{a = \frac{v_e}{m}\frac{\Delta m}{\Delta t} - g}$

$$\boldsymbol{a = \left(\frac{2.4 \times 10^3 m/s}{2.8 \times 10^6 kg} \times 1.4 \times 10^4 kg/s\right) - 9.8 m/s^2}$$

$$\boldsymbol{a = 2.20 m/s^2}$$

✓ **Rocket velocity, V is given by:**

$$\boldsymbol{V = v_e ln \frac{m_0}{m_f}}$$

Where **V** is rocket velocity, $\boldsymbol{v_e}$ is exhaust velocity, $\boldsymbol{m_0}$ is initial mass of rocket, $\boldsymbol{m_f}$ is final mass of rocket.

Factors affecting rocket acceleration

✓ The greater exhaust velocity, $\boldsymbol{v_e}$ of gases relative to the rocket, the greater acceleration.

✓ The faster the rocket burns its fuel, the greater its acceleration.

✓ The smaller the rocket's mass, the greater acceleration.

❖ **Air resistance**

✓ **Air resistance** is a force acting opposite to relative motion of any object moving with respect to surrounding fluid.

✓ It is a frictional force air exerts against a moving body.

✓ As an object moves, air resistance slows it down.

Air resistance is also known as **drag force**, F_D and it is given by:

$$\boldsymbol{F_D = \frac{1}{2}\rho V^2 C_D A}$$

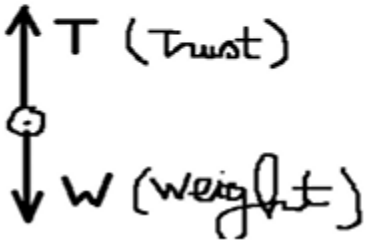
Where $\boldsymbol{F_D}$ is drag force or air resistance force, $\boldsymbol{\rho}$ is density of air, **V** is speed of rocket, $\boldsymbol{C_D}$ is drag coefficient, and **A** is cross sectional area.

Free body diagram on a rocket

A rocket is experienced by:

- ✓ **Trust force**: Force pushing up the rocket, and
- ✓ **Weight**: force pulling down the rocket

Then, the free diagram for the rocket looks like this;



Learning unit 2. APPLY STATIC EQUILIBRIUM AND ELASTICITY

2.1 Explanation of equilibrium conditions

Static equilibrium: is a state in which a system is stable and at rest.

To achieve static equilibrium, a system must have both rotational equilibrium and transitional equilibrium.

-Transitional equilibrium: is a state in which net force is equal to zero

-**Rotational equilibrium:** if the angular velocity is constant to mean that the net torque acting on the object must be zero.

2.1.1. Moment of the force

A moment of force (Torque): - is the turning effect of force

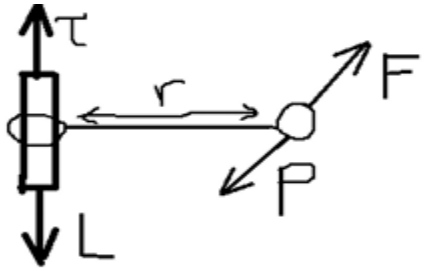
-is a measure of the rotational force applied to an object

The basic formula for calculating torque is:

$$\tau = F \cdot r \cdot \sin(\theta)$$

Where:

- τ = Torque (measured in Newton-meters, N·m)
- F = Applied force (measured in Newtons, N)
- r = The distance from the axis of rotation to the point where the force is applied (lever arm), measured in meters (m)
- θ = The angle between the force vector and the lever arm (measured in degrees or radians)



2.1.2 Necessary conditions for equilibrium of an object

➢ **First condition for equilibrium**

✓ The sum of all external forces acting on the body is zero,

$$\sum \boldsymbol{F = 0}$$

➢ **Second condition for equilibrium**

✓ The sum of all external torques from external forces is zero.

$$\sum \boldsymbol{\tau = 0}$$

2.1.3 Centre of gravity

The center of gravity of a body: is the point from which the whole weight of the body appears to act. It is also an important concept in determining stability of an object.

The center of gravity of a regular lamina or object is at its geometric center.

Three states of equilibrium of body

1. **Stable equilibrium:** if a body is displaced slightly from its equilibrium position, it experiences a force that tends to return it to its original position.

2. **Unstable equilibrium**: if a body is displaced slightly from its equilibrium position, it experiences a force that pushes it further away from its original position, moving the object further from equilibrium.

3. **Neutral equilibrium:** if a body is displaced slightly from its equilibrium position, it neither experiences a force that returns it to the original position nor a force that pushes it further away. The object stays in its new position after being displaced.

2.2 Application of elastic properties

2.2.1 Deformation of solids in terms of the concepts of stress and strain

-Stress is a quantity that describes the magnitude of forces that cause deformation.

Stress is generally defined as force per unit area.

$$Stress = \frac{F}{A}$$

The SI unit of stress is Pascal (Pa)

-Strain is the change in length divided by original length of object

$$Strain = \frac{\Delta L}{L_0}$$

Strain is dimensionless (no unit)

There are three (3) types of stress and strain:

- **Tensile stress:** change in length.

Then, the strain under tensile stress is called **tensile strain**.

- **Bulk stress:** change in volume.
  Then, the strain under bulk stress is called **bulk strain or volume strain.**
- **Shear stress:** change in geometry.
  Then, the strain under shear stress is called **shear strain.**

The ratio of stress and strain is called **elastic modulus** and SI unit is **the same as stress, (Pa).**

$$Elastic\ modulus = \frac{stress}{strain}$$

2.2.2. Three types of deformation and elastic modulus for each.

-Young's modulus: is the elastic modulus for tensile stress.

It is denoted as **Y**. **Young's modulus** is a ratio of tensile stress to tensile strain.

$$Y = \frac{Tensile\ stress}{Tensile\ strain} = \frac{F_{\downarrow}/A}{\Delta L/L_0} = \frac{F_{\downarrow}}{A} \times \frac{L_0}{\Delta L}$$

where **A** is area ΔL is change in length, $F\downarrow$ $A \times L0$ ΔL is deforming force, $L0$ is initial length.

-Bulk modulus: is the elastic modulus for bulk stress. It is denoted by **B**. It is a ratio of bulk stress to bulk strain.

$$B = \frac{Bulk\ stress}{Bulk\ strain} = -\frac{\Delta P}{\Delta V/V_0} = -\Delta P \frac{V_0}{\Delta V}$$

Note: The minus sign (-) in this equation is for consistency, to ensure that B is positive quantity.

- ✓ Note that minus sign (-) is necessary because an increase ΔP in pressure (positive quantity) always causes a decrease ΔV in volume, and decease in volume is a negative quantity.
- ✓ The reciprocal of the bulk modulus is called compressibility, **K** or,

$$K = \frac{1}{B} = -\frac{\Delta V/V_0}{\Delta P} = -\frac{\Delta V}{\Delta P V_0}$$

- ✓ The term compressibility is used in relation to fluids (gas and liquid).
- ✓ **Compressibility** is the change in volume of fluid per unit increase in pressure.

-Shear modulus: It is a ratio of shear stress to shear strain and is denoted as **S**.

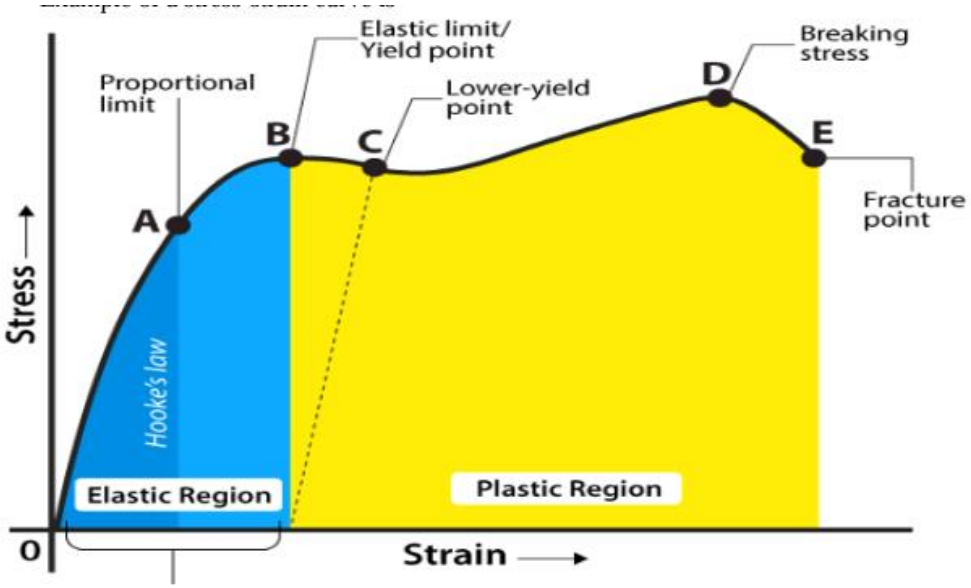
$$Shear\ strain = \frac{\Delta X}{L_0}$$

$$Shear\ stress = \frac{F!!}{A}$$

The shear stress is due to the force parallel to the surface $F!!$.

$$S = \frac{F!!/A}{\Delta X/L_0} = \frac{F!!}{A} \times \frac{L_0}{\Delta X}$$

2.2.3. Stress versus strain curve for an elastic solid



1. Proportional limit: It is a region in stress-strain curve that obey Hook's law.
In this limit, the stress-strain curve ratio gives us proportionality constant known as **Young's modulus.** The point 0A in the graph represent proportional limit.
2. Elastic limit: It is the point in the graph up to which the material returns to its original position when the load acting on it is completely removed.
Beyond this limit, the material does not return to its original position and plastic deformation starts to appear in it.
3. Yield point: It is defined as the point at which material starts to deform plastically.

- ✓ After yield point is pasted, permanent plastic deformation occurs.

- ✓ There are two yield point: **Upper yield point** and **Lower yield point**.

4. Ultimate stress point: It is a point that represents the maximum stress that material can endure before failure. Beyond this this point failure occurs.

5. Fracture or Breaking point: It is a point in the stress-strain curve at which the failure of the material takes place.

- ➢ **Hooke's law**

Hooke's law states that "the strain of material is proportional to the applied stress within the elastic limit of that material".

Mathematically, Hooke's law is commonly expressed as:

$$F = -KX$$

Where **F** is a force, **X** is extension length, **K** is constant of proportionality known as spring constant in N/m.

Example: A spring is displaced by 5cm and held in place with a force of 500N. What is the spring constant of the spring?

Ans: Given $F = 500N, X = 5cm = 0.05m$

Asked: K =?

Formula: $F = -KX,\ then\ !-K! = \frac{F}{X}$

$$K = \frac{500N}{0.05m} = 10000\ N/m$$

Learning unit 3: Analyze fluid mechanics

3.1. Description of pressure and its variation with depth

3.1.1. Pressure

-Pressure is perpendicular force per unit area.

It is a **stress** at point within confined fluid.

The SI unit of pressure is **Pascal (Pa).**

A. Pressure in fluid

Fluids: are substances that have no fixed shape and yield easily to external pressure.
Examples are gas and liquids

Pressure in fluids is given by,
P=Po+ρhg
Where **P** is pressure, P**o** is atmospheric pressure, ρ is density of fluids, h is height or depth and g is acceleration due to gravity.

If the atmospheric pressure, $P0 < 1$, pressure will be,
P =ρhg

B. Force and pressure

Force: is the pull or push action resulting in acceleration of an object.

Pressure: is a force acting upon a certain area and acted upon something perpendicular to its surface.

$$P = \frac{F}{A}$$

Where **P** is pressure, **F** is force, and **A** is area.

❖ **Difference between force and pressure**

Force	**Pressure**
Force is a pull or push action resulting in acceleration of an object.	Pressure is a force acting upon a certain area and acted upon something perpendicular to its surface
Its SI unit is Newton (N)	Its SI unit is Pascal (Pa)
Instrument used to measure force is called **Dynamometer**	Instrument used to measure force is called **Manometer**
Force is a vector quantity which means that has direction	Pressure is scalar quantity which means that no direction
Force can be acted upon the face, side, edges or vertices of the objects	Pressure only acts on surface or face of object.
Force change the speed and direction of an object.	Pressure cannot change the speed and direction of an object

C. A simple device for measuring the pressure exerted by a fluid

The instrument used to measure the pressure in fluid is called Manometer.

Manometer is a U-tube device used to measure pressure

3.1.2. Variation of pressure with depth

A. Variation of atmospheric pressure against altitude

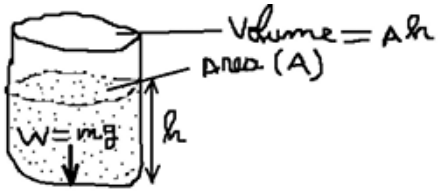
As altitude increases, the amount of gas molecules in the air decreases, the air becomes less dense than nearer to sea level.

This means that, atmospheric pressure is lower at higher altitude and it becomes higher at lower altitude.

B. Water pressure against depth

The pressure increases as depth increases.

This is due to the weight of water above the object in water and air above the water.



✓ Pressure exerted on the bottom by weight of fluid

$$P = \frac{W}{A} \quad \leftrightarrow \quad P = \frac{mg}{A}$$

✓ Mass of fluid from its volume and density,

$$\rho = \frac{m}{V} \quad \leftrightarrow \quad m = \rho V$$

✓ Volume of fluid is related to dimension of container,

$$V = Ah$$

Where **A** is cross sectional Area, **h** is depth

✓ **Combining the last two equations gives,**

$$m = \rho Ah$$

✓ Replace the value of mass, m in equation of pressure, P, we get,

$$P = \frac{\rho Ahg}{A}$$

A cancels, and rearranging the variables yields,

$$P = \rho gh$$

✓ This value is pressure due to the weight of fluid.

Example1: The dam is 500m wide, and the water is 40,0m deep at the dam.

a) What the average pressure on the dam due to the water?

b) Calculate the force exerted against the dam and compare it with the weight of water in the dam. (weight of water in dam, $W = 1.96 \times 10^{13} N$, density of water, $\rho = 10^3 kg/m^3$)

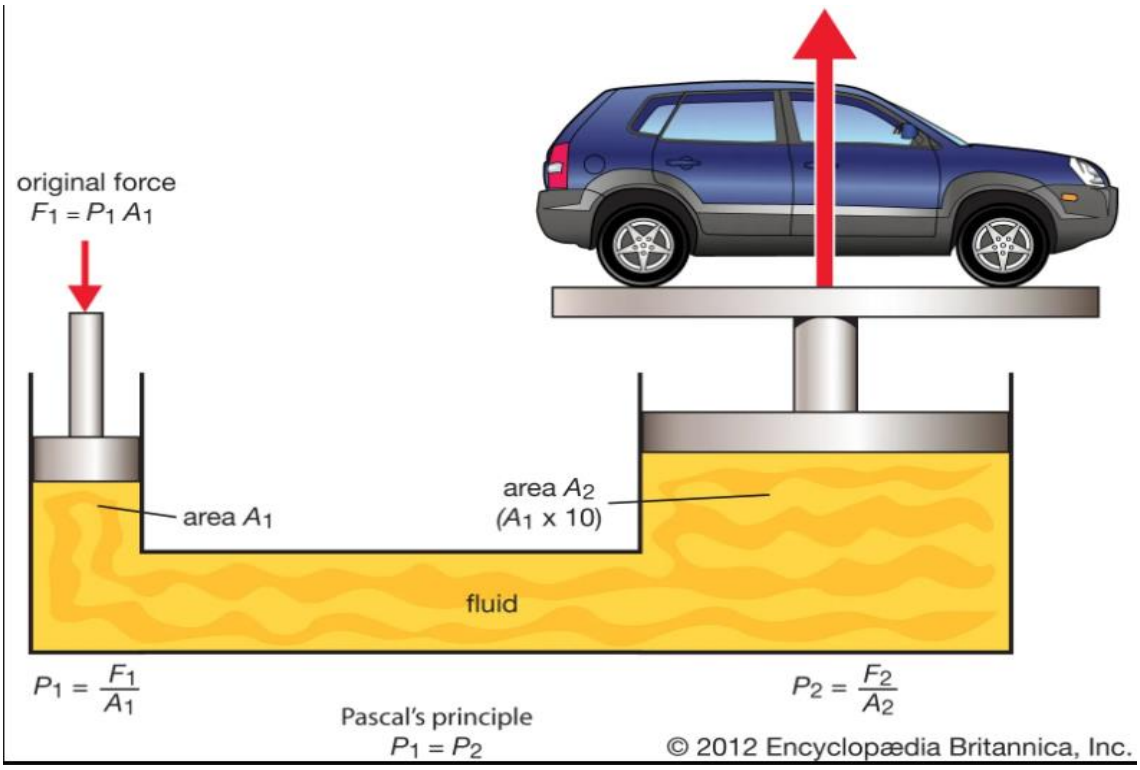
Example2: Calculate the depth below the surface of water at which the pressure due to the weight of water equals 1atm and density of water, ρw is 103 kg/m^3.

C. Pascal's principle

Pascal's Principle is also known as **Pascal's law**

According to Pascal's law, "Pressure or intensity of pressure at point in a static fluid will be equal in all directions". Or

Pascal's Principle states that "**The external static pressure applied on a confined liquid is distributed or transmitted evenly throughout the liquid in all directions".**



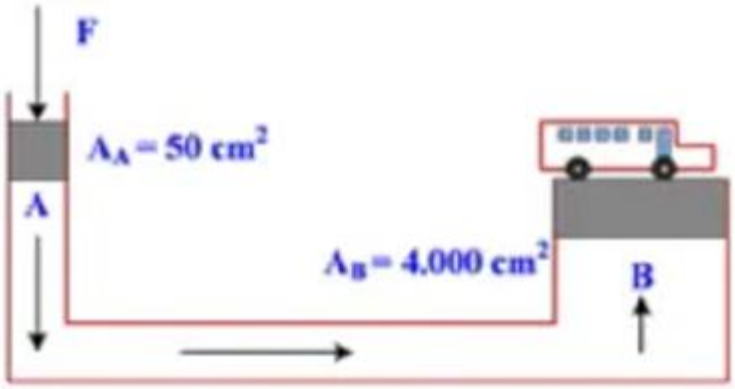
$$P_1 = P_2$$

$$\frac{F_1}{A_1} = \frac{F_2}{A_2}$$

$$\frac{F_1}{F_2} = \frac{A_1}{A_2}$$

Example1: Two pistons of a hydraulic lift have diameters of 60 cm and 5 cm. What is the force exerted by the larger piston when 50 N is placed on the smaller piston?

Example2: Car's weight is 1600N. What is the external force input force F?



Example3: A pressure of 2000Pa is transmitted throughout a liquid column due to a force being applied on a piston. If the piston the area of 0.1$m2$, what force is applied?

Applications of Pascal's Principle

1. Hydraulic Lift: It works based on the principle of equal pressure transmission throughout a liquid.

2. Hydraulic Brakes

3.1.3. Pressure measurements

a. Measurements of atmospheric pressure

Atmospheric pressure is measured in**:**

1. Atmosphere (atm)

2. Bar (bar)

3. Millimeters of mercury (mmHg)

4. Torr

$$1atm = 760torr = 760mmHg = 1.01325bar = 101325Pa$$

Note that:

- **Barometer** is an advice used to measure atmospheric pressure at a place.

- In a barometer, there is mercury column in which mercury moves up or down depending on

the atmospheric pressure at a place.

- It is graduated in millimeter and the unit used for measurements from a Barometer is mmHg.

$$1mmHg = 133.3224Pa$$
$$1atm = 760mmHg$$

Pressure measured by a Barometer is also known as **Barometric pressure (atmospheric**

pressure).

b. Absolute pressure and Gauge pressure

Absolute pressure: is a pressure of having no matter inside a space or perfect vacuum

The measurement of barometric pressure is example of absolute referenced pressure.

$$P_a = P_g + P_{atm}$$

Gauge pressure: It is measured in relation to ambient atmospheric pressure.

$$P_g = P_a - P_{atm}$$

Atmospheric pressure: Pressure measured by a Barometer.

$$P_{atm} = P_a - P_g$$

- **Difference between atmospheric pressure and gauge pressure**

- ✓ Atmospheric pressure is a measure of absolute pressure and is due to the weight of air molecules above the certain height relative to sea. It is increasing with decreasing altitude and decreasing with increasing altitude.
- ✓ Gauge pressure is additional pressure in a system relative to atmospheric pressure.

3.2. Apply Archimedes' Principle

3.2.1. Buoyant force

Buoyant force is an upward force exerted by any fluid upon a body placed in it.

It is an ability of something to float or rise in fluid.

$$Buoyant\ force = Weight\ of\ object\ in\ empty\ space - Weight\ of\ object\ immersed\ in\ fuid$$

$$F_B = \rho g V$$

Where F_B is Buoyant force applied on submerged object, ρ is density of fluid, V is volume of displaced fluid, g is acceleration of gravity.

$$Weight\ of\ displaced\ fluid = Weight\ of\ object\ in\ vacuum - Weight\ of\ object\ in\ fluid$$

$$Weight\ of\ displaced\ fluid = Buoyant\ force$$

3.2.2. Archimedes' Principle

Archimedes' principle states that "When a body is totally or partially immersed in a fluid it experiences an upthrust force equal to the weight of the fluid displaced".

Examples:

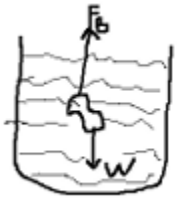
1. A ship that is launched sinks into the ocean until the weight of water it displaces is just equal

to its own weight.

2. A stone sink in the basin full of water, the weight of water displaced by stone is equal to the

weight of the stone.





$$F_B = m_f g$$

$$m_f: mass\ of\ fluid$$

$$W = F_B$$

3.3. Analyze fluid dynamics

3.3.1. Viscosity

-**Viscosity** is a measure of fluid's resistance to flow

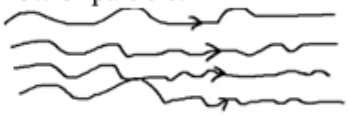
- It is a resistance of fluid (liquid or gas) to change in shape or movement of neighboring portions relative to one another.

a. Four assumptions of model of ideal fluid flow

1. Steady flow/ laminar flow: the velocity of fluid at each point remains constant in time. The particles move in straight line.



2. Unsteady flow/ Turbulent flow: Particles unlike straight line in their motion. It is irregular flow of particle.



3. Incompressible flow: Density of fluid remains constant

4. Irrotational flow: The motion of particles is in circular path but the rotation is not about the axis. There is no turbulence.

Turbulence: instability or disturbance

5. Non-viscous flow: No fluid friction.

---

b. Equation of continuity for fluids

Equation of continuity states that in the case of steady flow, the amount of fluid flowing past one point must be the same as the amount of fluid flowing past another, or mass flow rate is constant.

$$\rho_1 A_1 V_1 = \rho_2 A_2 V_2$$

Where $\boldsymbol{\rho}$ is density, $\boldsymbol{A}$ is cross sectional area, $\boldsymbol{V}$ is flow velocity of fluid, 1 and 2 indicate two

different region.

Example1: Suppose water is flowing through the pipe of diameter 1cm with flow velocity of

$2m/s$. If the pipe widens to a diameter of 3cm, what is the new flow rate?

Example2: suppose that a compressible gas is flowing through a pipe. In a region of the pipe with a cross sectional area of 0.02$m2$, it has a flow rate of 4$ms/$ and density of 2$kgm3/$. What is its density as it flows through another region of the same pipe with cross sectional area of 0.03$m2$ at velocity 1$ms/$?

Note that:

1. The volume flow rate (Q) is given by:

$$Q=AV$$

Where $\boldsymbol{Q}$ is volume flow rate, $\boldsymbol{A}$ is cross sectional area, and $\boldsymbol{V}$ is average velocity.

The IS unit of volume flow rate is $\boldsymbol{l/s}$ or $\boldsymbol{ml/min}$.

2. Mass flow rate ($\boldsymbol{m}$), is given by:

$$\dot{m} = \rho AV$$

- **Volume flow rate problem solving**

Example: The pipe with radius of 0.1m is used to drain water cylindrical tank with height of 3m and a diameter of 5m in less than one hour. How fast will the stream of water need to move through the pipe in m/s in order to fill of tank?

Ans: Volume of cylinder , $V = \pi r^2 h$

$$r = \frac{d}{2} = \frac{5m}{2} = 2.5m$$

$$V = 3.14 \times (2.5m)^2 \times 3m = 58.9m^3$$

Volume flow rate, $Q = AV$

Asked: To find the average velocity (V) of the pipe,

$$V = \frac{Q}{A}$$

Volume flow rate (Q), $Q = \frac{Volume}{Time}$

$$Time = 1h = 3600sec$$

$$Q = \frac{58.9m^3}{3600s} = 0.01636\, m^3/s$$

Area of drainage pipe (A), $A = \pi r^2$

$$A = 3.14 \times (o.1m)^2 = 0.0314m^2$$

Average velocity (V), $V = \frac{Q}{A}$

$$V = \frac{0.01636\, m^3/s}{0.0314m^2} = 0.52\, m/s = 52\, cm/s$$

Water must be forced through the pipe quick but plausible speed of about half meter per second to properly drain the tank.

3.3.3. Bernoulli's principle and its application

Bernoulli's principle states that "The total mechanical energy of the moving fluid comprising the gravitational potential energy of elevation, the energy associated with the fluid pressure and the kinetic energy of fluid motion, remains constant**".**

Bernoulli's principal formula

Bernoulli's principle formula is a relation between pressure, kinetic energy and gravitational potential energy of fluid in a container.

$$P + \frac{1}{2}\rho V^2 + \rho gh = constant$$

Where $\boldsymbol{P}$ is pressure exerted by fluid, $\boldsymbol{V}$ is velocity of fluid, $\boldsymbol{\rho}$ is density of fluid, $\boldsymbol{h}$ height of container.

At any point of pipe, the constant is the same,

$$P_1 + \frac{1}{2}\rho {V_1}^2 + \rho g h_1 = P_2 + \frac{1}{2}\rho {V_2}^2 + \rho g h_2$$

Bernoulli's equation at constant depth (h)

$$h_1 = h_2$$
$$P_1 + \frac{1}{2}\rho {V_1}^2 = P_2 + \frac{1}{2}\rho {V_2}^2$$

Example: Calculate the pressure in the hose whose absolute pressure is 1.01 × 105 $N\ m2$ speed of water in the hose increases from 1.96$m\ s$ and the density is 103 $kg\ m3$.

Ans: Pressure at point 2, $P_2 = 1.01 \times 10^5\ N/m^2$
Density of fluid, $\rho = 10^3\ kg/m^3$
Velocity of fluid at point 1, $V_1 = 1.96\ m/s$
Velocity of fluid at point 2, $V_1 = 25.5\ m/s$
Asked: P_1 =?
Bernoulli's equation:

$$P_1 + \frac{1}{2}\rho {V_1}^2 = P_2 + \frac{1}{2}\rho {V_2}^2$$

$$P_1 = P_2 + \frac{1}{2}\rho {V_2}^2 - \frac{1}{2}\rho {V_1}^2$$
$$P_1 = P_2 + \frac{1}{2}\rho\left({V_2}^2 - {V_1}^2\right)$$
$$P_1 = \left(1.01 \times 10^5\right) + \left(\frac{1}{2} \times 10^3 \times ((25.5)^2 - (1.96)^2)\right)$$
$$P_1 = 4.24 \times 10^5\ N/m^2$$

Application of Bernoulli's principle

1. **Airflight:**
   - ✓ To manufacture the wings of plane, you use Bernoulli's Principle
   - ✓ The top of wings is somewhat curved, while the bottom of wing is totally flat
   - ✓ While in the sky, air travels across both top and bottom concurrently.
   - ✓ Because both top part and bottom part of plane are designed differently this allows for the air on bottom to move slower, which creates more pressure on bottom and allows for the air on the top to move faster which creates less pressure.
   - ✓ This is what creates lift which allows planes to fly.
2. **Lift:** As you see on airplane, the wing moves up and down a little bit as it flies through the air.
3. **Baseball**
4. **Draft**
5. **Sailing**

Learning unit 4: Apply thermodynamics

LO4.1: Describe measurement of temperature of a body

4.1.1. Key terms

1.Internal energy (intrinsic energy): is the total energy of closed system.

It is the sum of potential energy of system and system's kinetic energy.

2. Surroundings: refer to everything outside the system that can interact with the system in some way, such as heat, work, or mass exchange

3. System: refers to the specific part of the universe that is under investigation. It could be a quantity of gas, a liquid, or any substance or collection of matter that is being analyzed

4. Boundary: is the surface that separates the system from its surroundings.

5. Extensive Properties: Extensive properties are those properties of a system that depend on the quantity or size of the material present in the system. Examples include mass, volume, energy, and entropy

6. Intensive Properties: Intensive properties are properties of a system that do not depend on the quantity or size of the material present in the system. They are inherent to the material itself. Examples include temperature, pressure, density, and specific volume

4.1.2 Heat Transfer and Quantity of heat

a. Heat and Temperature

1.Heat: is a form of energy that is transferred between systems or objects with different temperatures.

It flows from high temperature system to lower temperature system.

2.Temperature: It is a physical quantity that express hotness and coldness of the object.

It is expressed in several scales including Fahrenheit, Celsius, kelvin.

Heat

Heat is measured by **Thermometer**

Conversion of temperature scales (Celsius, kelvin, and Fahrenheit)

Conversion	Formula
Celsius to Fahrenheit	$°F = \frac{9}{5}(°C) + 32$
Kelvin to Fahrenheit	$°F = \frac{9}{5}(K - 273) + 32$
Fahrenheit to Kelvin	$K = \frac{5}{9}(°F - 32) + 273$
Fahrenheit to Celsius	$°C = \frac{5}{9}(°F - 32)$
Celsius to Kelvin	$K = °C + 273$
Kelvin to Celsius	$°C = K - 273$

NOTE: When you are counting the temperature in:

1. Celsius, you start from zero **(0℃)**
2. Fahrenheit, you start from **32℉**
3. Kelvin, you start from **273K**

   0℃ = 32℉ = 273K

Exercises:

1) Convert 100°C and 0°C into °F and K

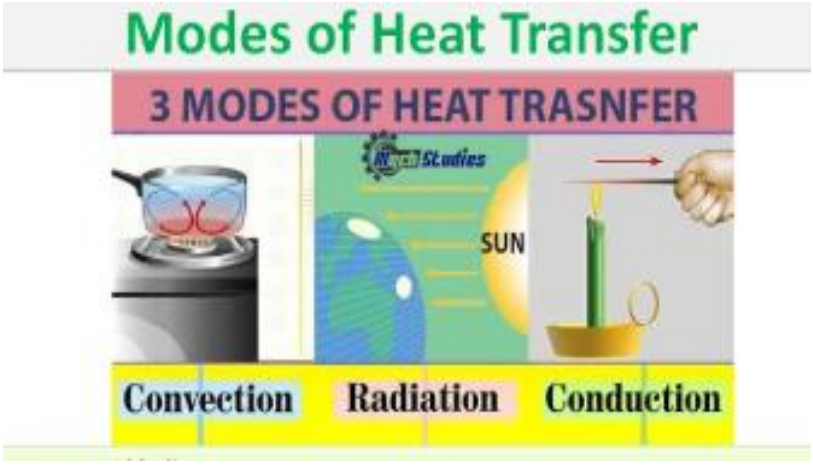
2) Convert 98.6°F into °C

b.Modes of heat transfer

1.Conduction: Heat is transferred between objects that are in direct contact with each other. Conduction is common in solids

2.Convection: Heat is transferred through the movement of fluids, like liquids and gases. Convection is common in liquids and gases

3.Radiation: Heat is transferred through electromagnetic waves. This can happen through empty space or any transparent medium



c. Heat capacity and Specific heat capacity

Heat capacity: defined as the amount of heat to be supplied to an object to produce a unit change in its temperature.

SI unit of heat capacity is joule per kelvin (J/K).

Specific heat capacity(c) of a substance: is the amount of heat that must be added to one unit of mass of the substance in order to cause an increase of one unit in temperature.

d. Thermal expansions

There are three types of expansion:

1. Linear expansion
2. Areal expansion
3. Volume expansion

1. Linear expansion: is the change in length due to heat.

Linear expansion formula is given as:

$$\frac{\Delta L}{L_0} = \alpha_L \Delta T$$

Where $L0$ is original length, L is expanded length, αL is length expansion coefficient, ΔL is change

in length, ΔT is change in temperature

2.Area expansion: is the change in area due to temperature change.

Area expansion formula is given as,

$$\frac{\Delta A}{A_0} = \alpha_A \Delta T$$

Where A is expanded area, $A0$ is original area, ΔA is change in area, ΔT is change in temperature, αA is area expansion coefficient.

Note that; $\alpha_A = 2\alpha_L$

3.Volume expansion: is the change in volume due to temperature

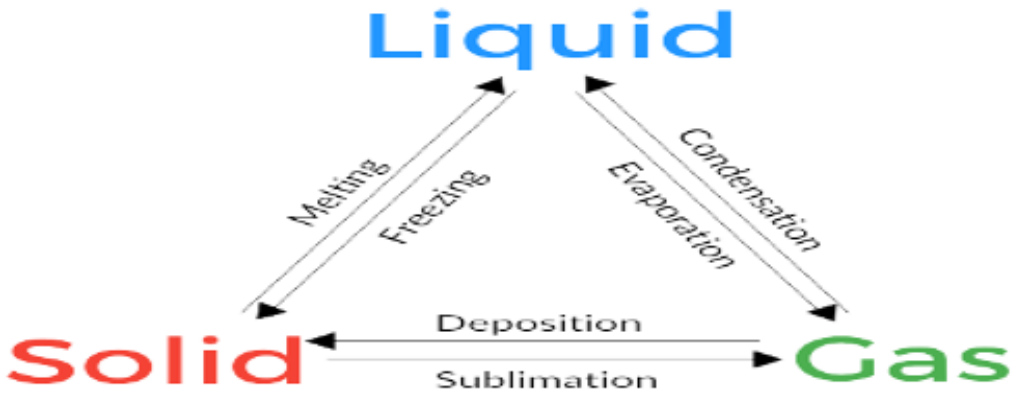
Volume expansion formula is given as,

$$\frac{\Delta V}{V_0} = \alpha_V \Delta T$$

Where $V0$ is original volume, V is expanded volume, ΔV is change in volume, ΔT is change in temperature, αV is volume expansion coefficient.

Note that, $\alpha_V = 2\alpha_A = 3\alpha_L$

e. <u>Transition of physical states</u>



- **Melting:** change from solid state to liquid state.
- **Freezing:** change from liquid state to solid state.
- **Condensation:** change from gas state to liquid state.
- **Evaporation:** change from liquid state to gas state.
- **Sublimation:** change from solid state to gas state.

4.2. Analyzing thermodynamic laws

a. Zeroth Law of Thermodynamics: "If two systems are each in thermal equilibrium with a third system, they are in thermal equilibrium with each other."

b. First Law of Thermodynamics: "that the change in the internal energy (ΔU) of a system is equal to the sum of the heat (Q) that flows across its boundaries and the work (W) done on the system by the surroundings."

$$\Delta U = Q - W$$

Where:

- ΔU is the change in internal energy of the system,
- Q is the heat added to the system,
- W is the work done by the system on its surroundings.

c. **Second Law of Thermodynamics:** "the spontaneous change for an irreversible process in an isolated system always proceeds in the direction

of increasing entropy. In other words, the entropy of any isolated system always increases."

4.3.Application of thermodynamic laws

-Refrigerators

-Heat engines

-Ventilators

-Power plant

Learning outcome 5: Examine effects of electric current flow in DC electric circuit

5.1. Description of a simple electric circuit

5.1.1 DC electric circuit

-DC electric circuit: is a circuit that contains the three basic components needed for an electric circuit to function.

Three basic components are

1. Source of voltage
2. Conductor
3. Resistor

-Electric current is movement of electrons or ions through an electrical conductor or space

It is measured in Ampere (A).

-Electrical resistance is the electrical quantity that measures how device or material reduces the electric current flow through it.

It is a ratio between the voltage applied to the current flowing through it

$$R = \frac{V}{I}$$

It is measured in Ohms (Ω)

-Voltage is difference in electric potential between two points

It is measured in Volt (V)

5.1.2. Measuring instruments used in simple DC electric circuit

1. Ammeter: Instrument used to measure current in circuit.

2. Voltmeter: Instrument used to measure potential difference or voltage between two points in

electrical circuit.

3. Ohmmeter: Electrical instrument used to measure resistance.

4. Multimeter: Electrical instrument used to measure more than one parameter like potential

difference, current, resistance, frequency, capacitance, etc.

5.1.3 Ohm's law

Ohm's law states that "The current flowing through a conductor between two points is directly proportional to the voltage across the two points and inversely proportional to the resistance of the conductor."

$$I \alpha V \quad \leftrightarrow \quad V = IR$$

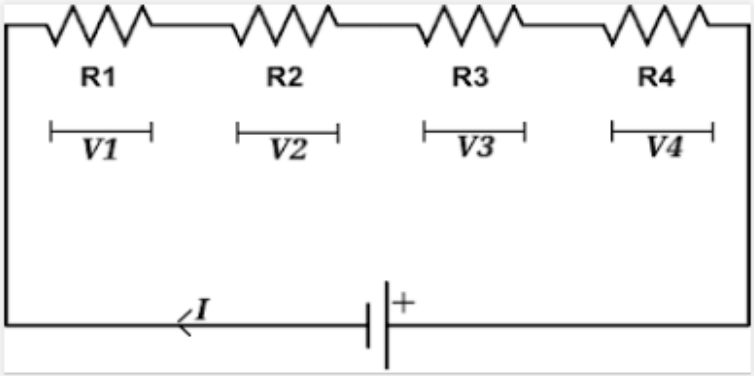
Where R: Resistance is constant

$$R = \frac{V}{I}$$

5.1.4. Combination of resistors

I. In series

Let $R1$, $R2$, $R3$…., Rn be the individual resistance which are connected in series, V is applied voltage and I is current passing in the circuit.



The current flow, **I** in this circuit is the same

i.e. $I = I_1 = I_2 = I_3 = \cdots = I_n$

The voltage in each resistor is different. Then, total voltage, V is given by

$$V_T = V_1 + V_2 + V_3 + \cdots + V_n$$

According to Ohm's law,

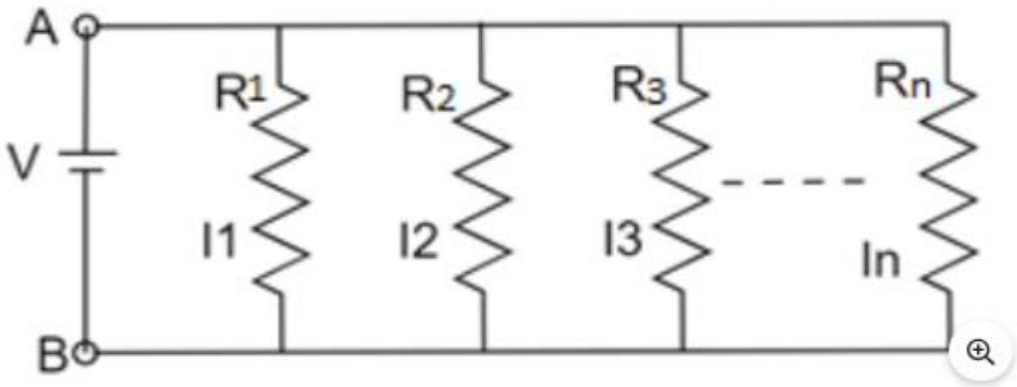
$$V_T = IR_T$$

$$V_1 = IR_1 \qquad V_3 = IR_3$$

$R_t = R1 + R2 + R3 + \cdots + R_n$

II. In parallel

Let $R1$, $R2$, $R3$…., Rn be the individual resistance which are connected in parallel, V is applied voltage and I is current passing through the circuit



The voltage across to each resistor is the same in parallel connection

$$V = V_1 = V_2 = V_3 = \cdots = V_n$$

The current across to each resistor in parallel connection is different.

$I_T = I_1 + I_2 + I_3 + \cdots + I_n$ where I_T is total current.

According to Ohm's law,

$$I_T = \frac{V}{R_T}$$

$$I_1 = \frac{V}{R_1} \qquad I_3 = \frac{V}{R_3}$$

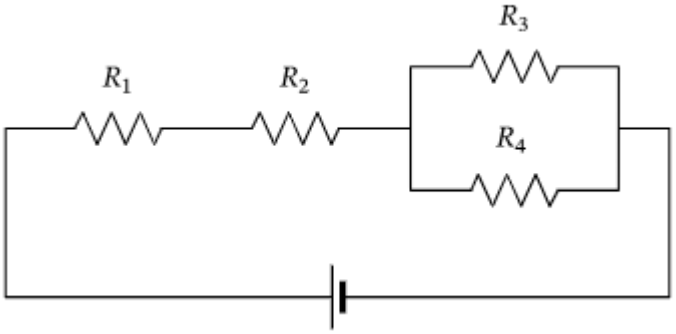
$$I_2 = \frac{V}{R_2} \qquad I_n = \frac{V}{R_n}$$

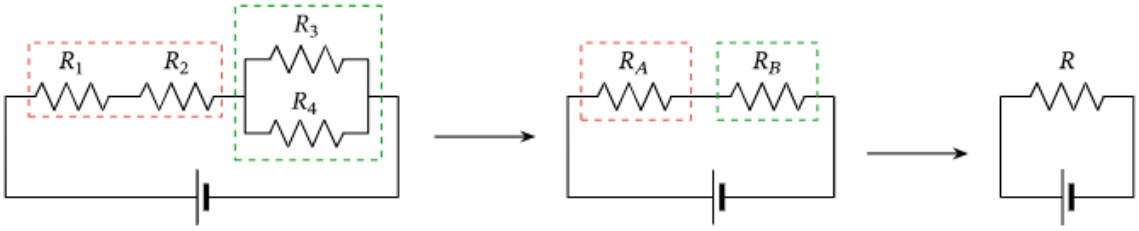
Resistance,

$$\frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \cdots + \frac{1}{R_n}$$

III. In mixed

Sometimes, the circuit may be neither pure series nor pure parallel, but a combination of series and parallel may come across as shown in the following figure





5.1.5. Electric cells

Electric cell is electrical power supply.

It converts stored **chemical energy** into **electrical potential energy**, allowing positive charges to flow from **positive terminal** to **negative terminal** one through external circuit.

Electromotive force, **emf**: energy supplied by battery or cell per Coulomb (Q) of charge passing through it.

$$emf = \frac{E}{Q} \quad or \quad e = \frac{E}{Q}$$

Where e is electromotive force (V), E is energy (J), Q is charge (Coulomb, C)

✓ emf formula can also be written as,

$$e = IR + Ir \quad or \quad e = V + Ir$$

Where e is electromotive force (V), I is current (A), R is load resistance (Ω), r is internal resistance of cell (Ω).

$$e = I(R + r)$$

5.2. Determine electric current, resistance and voltage in DC electric circuit

5.2.1. Key concepts

-Junction: Point at which at least three circuit paths meet

-Branch: It is a path connecting two junctions.

-Path: It is a continuous and closed loop. Path consists of conductors or wires and battery.

-Voltage supply: Voltage obtained from power source for operation of a circuit or device.

-Voltage drop: Reduction in an electrical circuit between the source and the load.

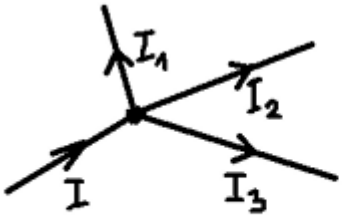
-Voltage gain: Difference between output signal voltage level in decibels and input signal

voltage level in decibels.

5.2.2 Kirchhoff's laws

1st Law of Kirchhoff: It is also known as Kirchhoff's current law (KCL).

It states that "The current flowing into a node (or junction) must be equal to the current flowing out of it".



$$I = I_1 + I_2 + I_3$$

Or, "The algebraic sum of currents meeting at electric junction in network is equal to zero". Assuming that currents entering in junction be positive and current out the junction be negative.

$$I + (-I_1) + (-I_2) + (-I_3) = 0$$

2nd Law of Kirchhoff: It is also known as Kirchhoff's law of voltage (KLV).

It states that "In any closed mesh or path, the algebraic sum of the products of current and resistance in each of the conductor plus the algebraic sum of the emf's in that path is equals to zero".

conductor plus the algebraic sum of the emf's in that path is equals to zero".

NOTES:

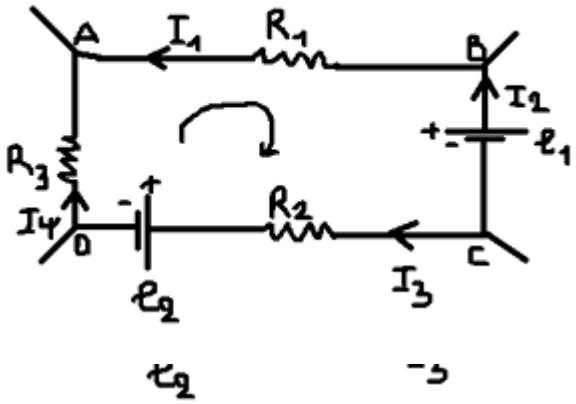
- We sign positive sign to voltage rise. i.e. if we start from the negative terminal of the battery

to its positive terminal.

- We sign negative sign to voltage drop. i.e. if we start from the positive terminal of the battery

to its negative terminal.

Consider a closed mesh ABCDA as shown in the following figure:



- ✓ Starting from point A and traveling through B to C to D to A.
- ✓ Signs or different voltage sources and voltage drops are as flows
- ✓ Let I_1, I_2, I_3, and I_4 are assumed direction through A-B, B-C, C-D, and D-A.
- ✓ I_1R_1 is a rise in voltage, so, **positive**.
- ✓ I_3R_2 is a voltage drop, so **negative**.
- ✓ I_4R_3 is a voltage drop, so **negative**.
- ✓ e_1 is a decrease in potential, so **negative**.
- ✓ e_2 is a rise in potential, so **positive**.

Now, apply KLV to mesh ABCDA

$$I_1R_1 - e_1 - I_3R_2 + e_2 - I_4R_3 = 0$$

$$Or \quad I_1R_1 - I_3R_2 - I_4R_3 = e_1 - e_2$$

5.3. Determine electric energy, work and power in DC electric circuit

5.3.1. Law of conservation of energy

It states that “Energy can neither be created nor destroyed only converted from one form of energy to another**”**

$$K_E + P_E = Constant$$

-Electric power: Type of kinetic energy caused by moving electric charges.

- It is energy derided as a result of movement of electrically charged particle.

- Electric power: Amount of energy produced per second.

-It is the rate at which electric energy is transferred.

-It is commonly expressed in megawatts (MW).

5.3.2. Calculations of energy and power in DC electric circuit

Electrical energy: It is capacity to do work and unit of work or energy in **Joule (J).**

$$Electrical\ energy = Power\ (W) \times Time\ (s)$$

$$1W = 1J/s$$

From this equation, we can find power.

$$Power = \frac{Electrical\ energy\ (E)}{Time\ (t)}$$

- **Electrical power**

$$Power = Voltage \times Current$$

$$P = VI \quad (1)$$

We know that $I = \frac{V}{R}$ **(2), Ohm's law**

Replace (2) in (1),

$$P = V.\frac{V}{R}$$

$$P = \frac{V^2}{R}$$

Again, $V = RI$, $P = RI.I$

$$P = RI^2$$

$$Hence, \quad P = \frac{V^2}{R} \quad or \quad P = RI^2$$

Where **R** is resistance

5.3.3. Effects associated with electric current in a circuit

A. Heating effect or Joule's effect

Joule's effect refers to the electrons moving in conductor cause the conductor to heat up.

-The amount of heat generated depends up on:

1. The amount of current flowing

2. Dimensions of the conductors

3. Type of conductor material used

B. Magnetic effect

-Whenever current flows in a conductor a magnetic field is set up around the conductor like an Extension of the insulation more about this later

-Increasing current increases the magnetic field.

- Switching the current off causes the magnetic field to collapse

C. Chemical effect

-When an electric current flows through a conducting liquid, the liquid separates into its chemical parts a process called electrolysis.

-Alternatively, if two metals are placed in conducting liquid they react chemically and produce voltage.

Learning unit 6. APPLY GEOMETRIC INSTRUMENT

6.1. Describe types of optical instruments

Optical instruments are all instruments used to aid vision.

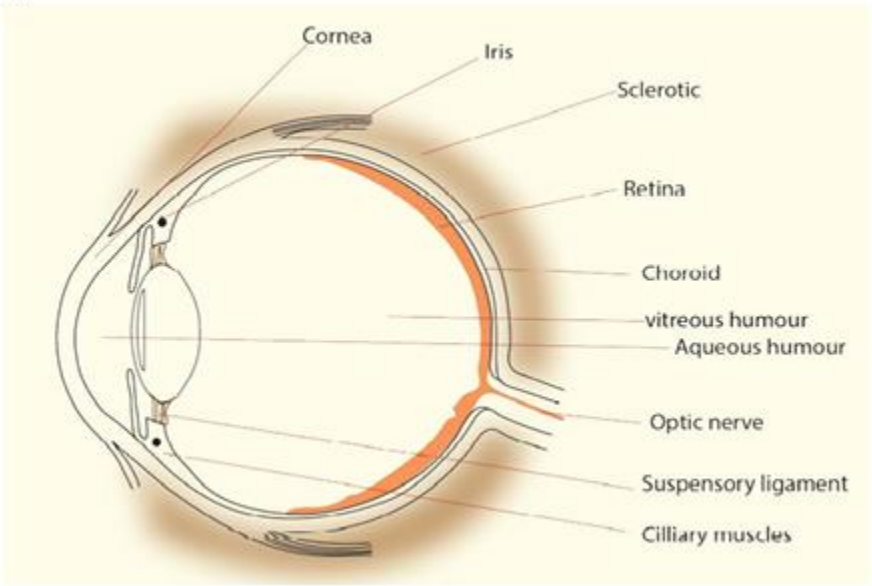
They are devices which process light wave to enhance an image for more clear view

6.1.1. Single lens optical instruments

1. Human eyes

Eye is biological instrument used to see objects at different distances

It uses a convex lens system to form small, inverted, and real image of an object in front of it.



Functions of the parts of the eye

-The cornea: It is made out of a fairly dense, jelly like material which provides protection for the eye, and seals in aqueous humor It also provides most of the power of the eye (59 Diopters), having about 46 Diopters. So, it provides most of the bending of light rays.

-Aqueous humor: This is a watery liquid that helps to keep the cornea in a rounded shape, similar to that of a lens.

-**Iris:** This controls the amount of light entering the eye.

-**Lens:** This is used to focus an image on the retina. It controls the bending of light rays by change of its shape, a process called accommodation, which is done by the ciliary muscles.

-**Ciliary muscles:** These control the thickness of the lens during focusing.

-**Retina:** This is the light sensitive part of the eye and it is where images are formed.

-**Vitreous humor**: This is a jerry like substance that helps the eye to keep its round shape.

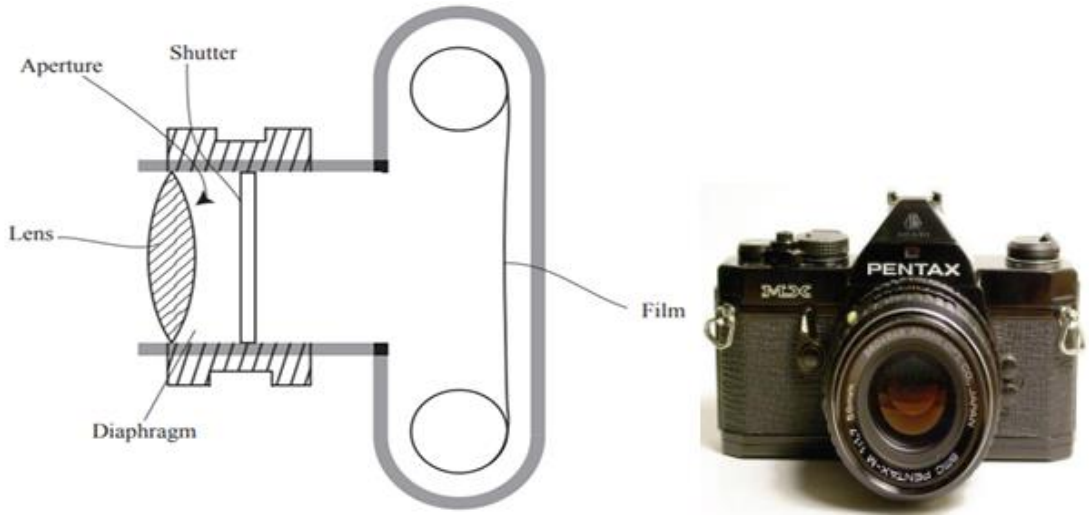
-**Yellow spot:** This is a small area on the retina where the sharpest image, that is, the finest detail can be seen.

-Optic nerve: This is the nerve that transmits images received by the retina to the brain for interpretation.

2. Magnifying glass

Camera that typically uses a mirror and prism system that permits the photographer to view through the lens and see exactly what we will be captured.

It uses the convex lens to form a small, inverted, real image on the film at the back



-The lens: focuses light from the object onto a light sensitive film. It is moved to and fro so that a sharp image is formed on the film.

-The diaphragm: is a set of sliding plates between the lens and the film. It controls the aperture (diameter) of a hole through which light passes.

-**The aperture**: controls the depth of field of the lens camera.

-The shutter: controls the exposure time of the film. It opens and closes quickly to let a small amount of light into the camera.

-**The exposure time**: affects the sharpness of the image. When the exposure time is short, the image is clear (sharp) but when it is long the image becomes blurred.

-**The film:** This is where the image is formed. It is kept in darkness until the shutter is opened. It is coated with light sensitive chemicals which are changed by the different shades and colours in the image.

6.1.2. Multi-lens optical instruments

Multiple lenses are devices like telescopes and microscopes which use multiple lenses to create images

1. Telescope

- It is an optical instrument using lenses, curved mirrors, or combination of both to observe

distant objects.

- It is also defined as various **devices used to observe distant object by emission, absorption, or reflection of electromagnetic radiation.**

2. Projector

- It consists of illumination system and projection lens.

- Illumination system consists of lamp, concave reflector and the condenser.

-Illuminant is either a carbon electric arc or quartz lamp to give small but very high intensity source of light in order to make image brighter.

6.2. Use of optical instruments

1.Uses of microscopes

- Microscopes used to observe smallest object that are at cellular level.

- Microscope allows scientists to see the shape of cell, its nucleus, mitochondria, and other organelles.

2. Use of telescope

- It is used to form magnified images of distinct objects.

6.3. Correction of optical aberrations

1. Spherical aberration

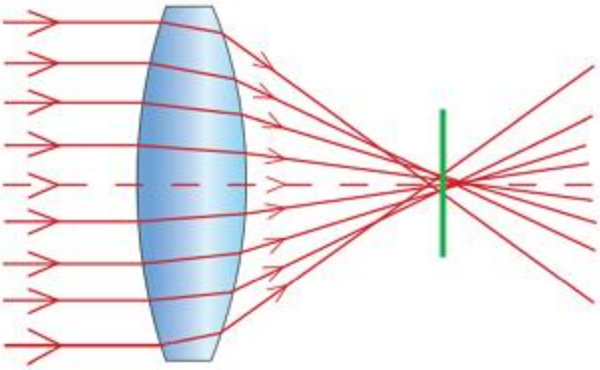
-This arises in lenses of large aperture when a wide beam of light incident on the lens, not all

rays are brought to one focus.

- As a result, the image of object becomes distorted.

-The defect is due to the fact that the focal length of lens for rays far from the principal axis are less than for rays closer to property of spherical surface and as result, they converge to a point closer to the lens.

a. **Lens cannot focus a single color of light**

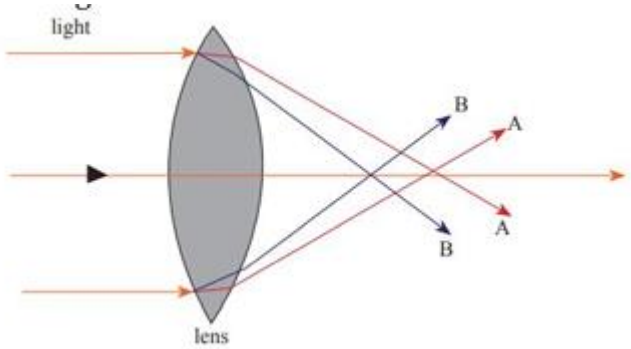


Correction: This defect can be minimized (reduced) by surrounding the lens with an aperture disc having a hole in the middle so that rays fall on the lens at a point closer to its principal axis.

2. Chromatic aberration

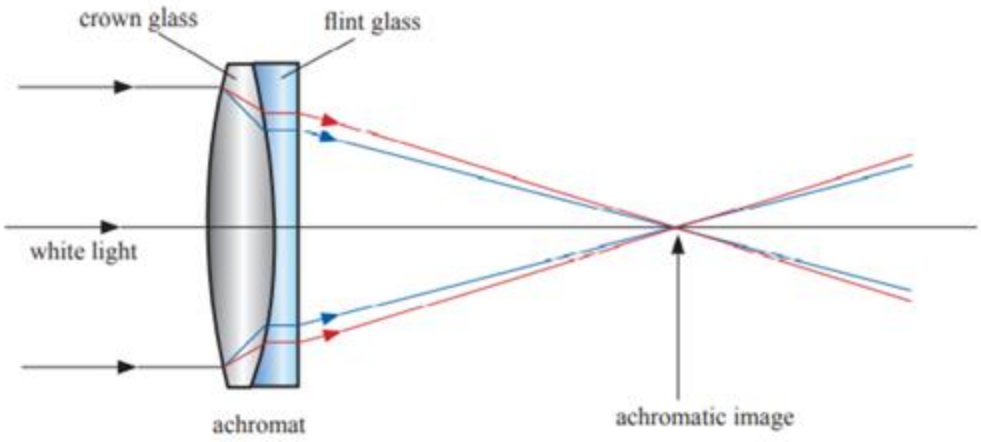
-It occurs when white light from object falls on a lens and splits it into its component colors.

-These colors separate and converge to different foci, and this results into an image with colored edges.



Simply**, lens do not focus the various color at single point**.

Correction: Chromatic aberration can be minimized by using an achromatic lens called an achromatic doublet.



3.Astigmatism aberration

-This is the defect that occurs if the curvature of the cornea varies in different directions so that rays in different planes from an object are focused in different positions by the eye and the image is distorted.

-A person suffering from astigmatism sees one set of lines more sharply than others.

Correction: This defect is corrected by wearing corrected lenses. These help to bend the incoming rays to correct for irregular refraction.

END